

Islam Ahmed Abdel moneim

Electronics and Communications Engineer/Industrial Automation Engineer /BMS Engineer

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Summary

Electrical Engineer specialized in PLC programming, automation systems, BMS, and embedded systems. Experienced in Siemens PLCs, SCADA, HMI design, Arduino/ESP development, and industrial control systems. Focused on automation, energy efficiency, and industrial process optimization.

Technical Skills

- **PLC & Automation:** Siemens S7-1200/1500, Allen-Bradley, Ladder Logic, SCADA, HMI, Factory I/O
- **Motor Drives:** AC/DC/Stepper Motor Control
- **Hardware:** STM32, ESP32, PCB Design, Embedded Systems, Control Panels
- **HVAC & Building Management Systems (BMS):** HVAC Systems, Chillers, FCU, Pumps Control, Monitoring & Automation
- **Software:** MATLAB, AutoCAD, Proteus, Arduino IDE, C/C++
- **Control Systems:** PID, Root Locus, Bode Plot, Stability Analysis

Work Experience

BMS Engineer — MEVAC Company

May 2025 – Present

- Managing BMS systems, SCADA monitoring and control, HVAC, Chillers and utility automation.
- Troubleshooting communication and sensor issues.
- Optimizing system performance and energy efficiency.

Automation Engineer — Diamond Automation

Jan 2025 – May 2025

- Developed PLC automation solutions and HMI interfaces.
- Designed and troubleshot industrial control panels.
- Implemented SCADA systems for remote monitoring and control.

Projects

- **PLC-Based Conveyor Belt Automation System**
Designed and programmed a Siemens PLC-based conveyor automation system, improving operational efficiency and reducing manual operation.
- **Smart Home Automation System**
Developed an IoT-based smart home system using ESP32, MQTT, and Firebase with smart monitoring and energy-saving control.
- **Printing Line Automation Project – RioPack**
Programmed and commissioned a 9-stage, 9-color printing line for large roll printing using Fatek PLC and SCADA systems.
- **Jigger Machine Automation Upgrade**
Converted manual jigger machines into fully automatic systems using tension control with dual motors, dual VSDs, Fatek PLC, and HMI.
- **Agricultural Incubator Control System**
Designed and implemented an automated agricultural incubator system for the Ministry of Agriculture using Siemens PLC.
- **BMS Project Contribution – MEVAC & Atech**
Contributed to Building Management System (BMS) projects using Schneider systems for HVAC, Chillers monitoring and control.

Education

Bachelor of Science in Electrical Engineering

Higher Technological Institute (HTI), Sharkia, Egypt — 2025

GPA: 3.5 / 4.00 | Graduation Project: Wireless Patient Health Monitoring System

Training & Certifications

- ITI IoT Internship — 2024
- Classic Control, PLC, Motor Drives & Panel Building — AH Consulting Group, 2024
- Electrical Protection & Power Systems Training — North Cairo Power Plant, 2022

Languages

- Arabic: Native
- English: Professional Working Proficiency